US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250283665
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Hammerich; Jörg

CALCINED CLAY COOLER

Abstract

A calcined clay cooler—that includes a rotatably supported clay conduit having a clay conduit wall (enclosing a clay conduit volume) with a clay inlet and a clay outlet located downstream from the clay inlet—provides an improved coolant and incurs reduced operating costs, if the clay conduit wall includes at least one coolant channel with a channel wall enclosing a channel volume and defining a channel inlet and a channel outlet.

Inventors:	**Hammerich; Jörg (Neustadt, DE)**
Applicant:	**ALITE GmbH** (Neustadt, DE)
Family ID:	**83004926**
Appl. No.:	**19/052747**
Filed:	**February 13, 2025**

Foreign Application Priority Data

EP	22191273.6	Aug. 19, 2022

Related U.S. Application Data

parent WO continuation PCT/EP2023/072359 20230814 PENDING child US 19052747

Publication Classification

Int. Cl.: **F28C3/04** (20060101); **C04B7/47** (20060101)

U.S. Cl.:

CPC **F28C3/04** (20130101); **C04B7/47** (20130101);

Background/Summary

BACKGROUND

1. Field of the Invention

[0001] The invention relates to the manufacture of pozzolanic or hydraulic materials for substitution of cement clinker, e.g., in Portland Cement manufacture. In more detail, the invention relates to a calcined clay cooler. The calcined clay cooler has an axis of rotation and at least one clay conduit. The clay conduit(s) may each have a clay conduit wall that encloses a clay conduit volume with an upstream clay inlet and a downstream clay outlet. The clay conduits are rotatably supported to rotate relative to the axis of rotation.

2. Description of the Related Art

[0002] Cement clinker is an approved concrete component, but its production is associated with significant CO.sub.2-emissions. To reduce the amount of released CO.sub.2 per ton of concrete it has been suggested to replace at least a portion of the clinker by pozzolanic-like materials. Examples are natural pozzolan and fly ash. Further, latent hydraulic materials such as ground granulated blast furnace slag have as well been suggested as a clinker replacing material. Unfortunately, the availability of these materials is limited and hence so called artificial pozzolans have been suggested. These artificial pozzolans can be manufactured by calcining clays, usually between 600° C. and 800° C. (see e.g., Rodrigo Fernandez, Fernando Martirena and Karen L. Scrivener: *The origin of the pozzolanic activity of calcined clay minerals: A comparison between kaolinite, illite and montmorillonite*; Cement and Concrete Research, Vol. 41, 2011, p 113-122). Industrial scale experiments have as well been reported by Tobias Danner, *Reactivity of Calcined Clays*, PhD-Thesis 2013, Trondheim University (ISBN 978-82-471-4552-4). Presently, in industrial scale, the clays are heated ('calcined', see below) by a clay furnace and subsequently cooled by means of calcined clay coolers. Such calcined clay coolers are rotating drums with a spray cooled drum shell. Water as coolant is sprayed on the peripheral surface of the drum (Luiz Felipe de Pinho and Roberto Campos: Claying it all on the line; World Cement, 2021, Vol. 4, p. 35-38, see last figures), while the calcined clay is transported in the rotating drum to the cooler outlet.

[0003] U.S. Pat. No. 11,530,881B2 suggests a bulk material cooler with open transport tubes for transporting material to be cooled indirectly. The transport tubes are arranged about an axis of rotation and are each adapted to be filled jointly via a conical filling region with material to be cooled. Each transport tube is arranged concentrically within a respective cooling tube. The cooling tubes may be connected to a single coolant tube via supply tubes which may be arranged e.g. in a spoke-like manner, wherein the cooling medium is fed into the coolant tube and flows via the supply tubes into the cooling tubes.

[0004] These burnt clays are referred to as "calcined clays", although calcination in its literal sense, being a thermal decomposition of CaCO.sub.3—, does not take place. It is assumed that that the term "calcination" is used simply because of the analogy to the cement clinker process: Limestone is rendered hydraulic by heating it above the calcination temperature and clay is rendered pozzolanic by heating it. Both materials can so to speak be 'activated' for the purpose of being used in the concrete industry. In other words, the term calcined clays simply refers to a clay that has been subjected to a heat treatment for rendering the clay pozzolanic and/or hydraulic.

SUMMARY

[0005] The problem to be solved by the invention and its embodiments is to improve cooling of calcined clay being provided to the cooler from a clay furnace, e.g., from a rotary clay kiln.

[0006] The calcined clay cooler has an axis of rotation and at least one clay conduit. The at least one clay conduit may preferably be supported to rotate around the axis of rotation and each of the at least one clay conduit(s) may have a separate conduit wall enclosing a separate conduit volume.

The at least one conduit volume may be accessible by an upstream clay inlet and a downstream clay outlet. In other words, the conduit volume has a clay inlet and a clay outlet, which may be defined by corresponding openings in the conduit wall. Although a single clay conduit may be sufficient, two or more clay conduits are preferred. Each of the at least one clay conduits may preferably be rotatably supported to rotate relative to the same axis of rotation. Of course, the calcined clay cooler may as well include non-rotating components, e.g., a support base providing a rotational invariant surface (or a section thereof) supporting at least indirectly the at least one clay conduit. The at least one support base may rotatably support at least one roller, preferably at least two rollers. The at least one roller may support the at least one clay conduit, at least indirectly, for example via at least one support ring. For example, the at least one clay conduit may be attached to a support wheel. At least two support wheels may be attached to each other by a support shell. On the peripheral surface of the support shell may be a tire ring. The tire ring may be rotatably supported relative to the support based by the at least one roller, preferably by at least two rollers.
[0007] The at least one clay conduit may be a part of rotating portion of the calcined clay cooler. The rotating portion may be rotatably supported by static portion, wherein the term static only implies that not the entire static portion moves. Of course, the static portion may include, e.g., rollers or the like, which allow to rotatably support the rotating portion. A drive may be coupled to the rotating portion, e.g., via an optional transmission, to drive the rotating portion.
[0008] Preferably, each of the clay conduit walls includes at least one coolant channel with a channel wall enclosing a channel volume and defining a channel inlet and a channel outlet. These coolant channels enable to effectively cool calcined clay being conveyed from the clay inlet to the clay outlet of the clay conduit, while reducing the coolant consumption and enabling to recuperate heat being removed from the calcined clay. The rotation of the clay conduit during said cooling enhances the thermal energy transfer from the calcined clay via the conduit wall to the coolant in the coolant channel. The coolant may be water or another fluid, wherein ‘fluid’ means that the coolant is in a fluid phase (i.e., in a liquid or gaseous phase) under the operating conditions (pressure and temperature) of the calcined clay cooler. The term fluid as well includes bulk materials, as these can be fluidized or flow similar to fluids if the slide-angle is exceeded.
[0009] Preferably, the calcined clay cooler has more than just a single clay conduit, e.g., two, three, four, five, six, seven, eight, nine or ten clay conduits. Theoretically, there is no limit for the number n of clay conduits, but if we have to define a maximum for formal and/or legal grounds one may consider that 800 or 1000 clay conduits are a maximum ($n \leq 1000$). The exact maximum depends on a number of factors. One factor is the diameter of the calcined clay cooler, another is the particle size of the hot clay to be cooled down, the tendency to agglomerate and as well the viscosity of the coolant. Generally, increasing the number of clay conduits increases the heat transfer surface separating the clay and the coolant and hence increases the heat transfer. On the other hand, increasing the number of conduits while maintaining the dimension of clay cooler essentially constant decreases the conduit diameter. The limit for increasing the number clay conduits is thus obtained if the free diameter of the clay conduit is so small that the burnt clay is not reliably transported by the clay conduit.
[0010] For a given total clay flow through the clay cooler, an increase of the number of clay conduits provides an increase of the surface via which a heat exchange may take place. Thus, the cooler can be dimensioned shorter (for a given calcined clay flow) while not significantly increasing the diameter of the trajectory being defined by the rotation of the clay conduit(s). Further, heat losses can be reduced (i.e., heat recuperation is improved) as well as the footprint of the calcined clay cooler. The latter is an important advantage when retrofitting existing plants. A preferred number n of conduits may be in between four and eight ($4 \leq n \leq 8$). In this range, the conduits are still accessible from the inside by humans while the ratio of the peripheral surface to the volume of the clay conduits is reasonably high to promote a high heat transfer rate and a short length of the calcined clay cooler. Increasing the number n of conduits is possible and further

enhances heat transfer from the calcined clay to the coolant, but has the disadvantage that manufacture, maintenance and inspection are more complicated.
[0011] The clay conduit(s) may preferably be arranged symmetrically with respect to the rotational axis. This symmetric arrangement minimizes static imbalances and homogenizes the torque required to rotate the rotating portion. Arranged symmetrically means in this context that in a preferred example a rotation of clay conduits around the rotational axis by
[00001]$\frac{2}{n} = \frac{360^\circ}{n}$
maps the respective clay conduit onto its neighbored clay conduit, wherein n is the number of clay conduits, assuming the centers of the clay conduits have the same radial distance to the rotational axis. In practical realizations deviations from the preferred angle
[00002]$\frac{2}{n}$
can be expected. In case margins for these deviations have to be defined, we would suggest a margin of ±30°, ±20°, ±15°, ±10°, ±5°, ±1°, wherein smaller margins are preferred. In another example there exists for each first clay conduit at least one second clay conduit onto which it can be mapped by rotating the first clay conduit by
[00003]$\frac{2}{m+1}$
wherein m is the number of second clay conduits. If the number of second clay conduits is greater than 1 (m≥2) than another rotation by
[00004]$\frac{2}{m+1}$
maps the first clay conduit onto the position of another second clay conduit. This can be repeated until the first clay conduit is mapped onto its initial azimuthal position.
[0012] In a preferred example, the calcined clay cooler may further include an inlet section. The inlet section may include an inlet duct with a duct inlet. Further the inlet section may include a number of duct outlets. Each of the duct outlets may preferably be in fluid communication with the duct inlet. The number of duct outlets preferably corresponds to the number clay conduits. For example, if the number of clay conduits is four, the inlet section may preferably have four duct outlets. Each of the duct outlets may be connected to the clay inlet of the corresponding clay conduit. For example, the first duct outlet may be connected to the first of the number of clay inlets, the second duct outlet may be connected to the second clay inlet and so forth. More generally, the i.sup.th duct outlet may be connected to the i.sup.th clay inlet, wherein i is an integer with 1≤i≤n and n is the number of clay conduits. In the example of n=2 the inlet section may include a duct with a bifurcation connecting the duct inlet with each of the clay inlets. Or more generally the duct splits up into a number of n duct sections. As already apparent, in a preferred example, the inlet section may be a separate duct that may be in fluid communication with the clay inlets of the clay conduits. It may bi- or multifurcate or simply end as a single duct right in front of the clay inlets of the clay conduits. The inlet section and the clay conduits can be attached to each other at least indirectly, e.g. via the support shell, thereby ensuring that the inlet section rotates with the clay conduits. The inlet section allows to distribute hot calcined clay from a single outlet of a clay kiln to the number of n clay conduits. Further, the inlet section provides for and is thus configured for a homogenization of the material to be cooled (e.g. calcined clay), which results in an even distribution of the material to the clay inlets of the respective clay conduits.
[0013] The inner surface of the inlet section may preferably be protected (and/or provided) by refractory. The inlet section is not necessarily actively cooled, but at least a portion of it may be in thermal contact with a coolant flow through an optional inlet section coolant channel.
[0014] Preferably, a connection chute having a connection chute end may extend into the duct inlet. A ring gap between the duct inlet and the chute end may be preferred and may allow the inlet section to rotate relative to the connection chute. The ring gap may be sealed by a gasket. In a preferred example, at least one fluid line may have at least one nozzle opening facing toward the ring gap and/or ending in the ring gap and/or in an enclosure of the ring gap. In other words, the nozzle opening is preferably configured to provide a fluid flow into the ring gap. In addition or

alternatively, the pressure in the inlet section may optionally be below ambient pressure. Thereby fluid, e.g. air may be sucked into the clay conduit(s) via the inlet section and further enhance cooling of the calcined clay. Other example fluids can be steam and/or carbon dioxide (CO.sub.2), nitrogen (N.sub.2), as apparent, these may preferably be provided to the ring gap via the at least one fluid line. Sealing the gap with a fluid and/or a gasket has the advantage that less hot gas being transported with the calcined clay escapes via the ring gap into the environment, which would result in an energy loss and as well in an unpleasant odor.

[0015] In addition or alternatively, the chute and/or an inlet section may include a water injection lance, allowing to treat the still hot calcined material with water and/or steam. For example, a lance may be attached to the chute and end in the inlet section or be elsewise configured to provide a fluid flow (e.g. a water and/or a steam flow) via at least one lance nozzle into the inlet section.

[0016] In a particularly preferred example, the coolant channel volume may be limited by the clay conduit wall and by a channel wall. During operation, this measure provides for a direct heat exchange between a coolant flowing through the coolant channel and the conduit wall, being particularly efficient. Further, the amount of wall material can be reduced, as the conduit wall serves as well to delimit the coolant channel volume. In a preferred example, the channel wall encloses at least a portion of the conduit wall, preferably the entire conduit wall. In this case, in a cross-sectional view at least a portion of the channel wall may be considered as being a ring surrounding the conduit wall. Each of these measures allows to increase the heat transfer between the coolant and the conduit wall and thus between the calcined clay and the coolant. For a given calcined clay flow the footprint of the calcined clay cooler can be further reduced, while increasing the amount of heat being recuperated.

[0017] For example, the channel wall may support the conduit wall radially and/or anti-radially, while allowing the conduit wall to float in the axial direction. Thereby, stress due to different axial heat expansions of the channel wall and the conduit wall can be reduced.

[0018] As usual, “radially” implies that a vector is pointing away from the corresponding axis. Accordingly, “anti-radially” implies that a vector points towards the corresponding axis. This means that the start point and the end point of a radial vector as well as of an anti-radial vector differ only in their distance from the corresponding axis. The azimuthal coordinates and the axial coordinates are identical. Unless otherwise defined, the corresponding axis of a conduit may be the conduit axis. In case of non-straight conduits, one may consider the cross sections and consider the center of the conduit in a cross section as starting point or end point of the radial or the anti-radial direction, respectively.

[0019] Preferably, the calcined clay cooler includes at least one support profile with a first leg, a middle leg and a second leg. As implied by the term, the middle leg may be in between of the first leg and second leg and defines an offset between the first leg and the second leg. The first leg may be attached to the peripheral surface of the conduit wall or to the inner surface of the channel wall. The second leg may hold the corresponding other wall in its radial position. For example, the second leg may have a plain bearing surface contacting the respective other of the peripheral surface of the conduit wall and an inner surface of the channel wall. Thus, if the first leg is (firmly) attached to the peripheral surface of the conduit wall the second leg's plain bearing surface may contact the inner surface of the channel wall. Alternatively, the first leg may be (firmly) attached to the inner surface of the channel wall and the second leg's plain bearing surface may contact the peripheral surface of the conduit wall. The middle leg may provide for a radial offset between the surface of the first leg being (firmly) attached and the plain bearing surface. Preferably, the calcined clay cooler includes as least three of these support profiles per clay conduit, wherein these support profiles may preferably be distributed preferably at least essentially evenly with respect to the azimuthal direction (e.g., evenly within ±45°) to thereby define the position of the corresponding clay conduit in the radial direction. The middle leg of the support profile may be of an elastic material, e.g., steel or the like, and hence compensate for different radial thermal expansions (and contractions) of the conduit wall and the channel wall, while maintaining the two walls in their relative position to each other. In an example, the two walls can be maintained at least essentially concentric while still allowing for a compensation of axial stress, e.g., due different thermal expansions.

[0020] Only to provide an example, the first leg may be attached to the corresponding wall by welding, brazing, riveting, or gluing or may even be unitary with the wall to which it may be attached.

[0021] Preferably, the calcined clay cooler includes a coolant line, wherein the coolant line extends parallel to, preferably concentric with, or more generally along the rotational axis. The coolant line may include a connection port in the downstream half of the calcined clay cooler and may be fluidly connected with the upstream end of the at least one coolant channel. The coolant line allows to locate rotary couplings in the colder end of the calcined clay cooler, thereby reducing the thermal stress. The construction can be cheaper and at the same time more durable and easier to service.

[0022] The recuperated exergy can be increased, and the required length of the calcined clay cooler can be minimized, by providing a counterflow of the coolant and the calcined clay. However, this may require very expensive materials in the upstream end of the conduits. This can be avoided by parallel flow of the coolant and the calcined clay. In a preferred example, the coolant channel(s) has (have) two sections, a first upstream section and a second downstream section, wherein upstream and downstream references to the flow direction of the calcined clay. In other words, the upstream section may be closer the to the clay inlet than the downstream section of the coolant channel and the downstream section of the coolant channel may be closer to the downstream clay outlet. First, the cold coolant may be provided to the upstream section, to thereby maintain the conduit wall below a given temperature. In the upstream section(s) of the coolant channel(s) the coolant may flow parallel or antiparallel to the calcined clay in the clay conduit. From an end of the upstream section, the preheated coolant may be provided to the downstream end of the second section. From there the coolant may flow towards an upstream end of the second portion of the coolant channel(s), i.e., in a counterflow with the calcined clay in the corresponding duct. This configuration may keep the upstream portion of the conduit wall sufficiently cold, while still providing for an increase of the recuperated exergy. As already apparent, upstream and downstream reference herein only to the conveying direction of the calcined clay.

[0023] For example, the calcined clay cooler may include a coolant cooler with a hot coolant inlet and a cold coolant outlet, wherein the coolant channel inlet may be connected with the cold coolant outlet and the channel outlet is connected with the hot coolant inlet. The coolant cooler may be a heat exchanger being configured to withdraw heat from the coolant and to provide this heat to a heat carrier, e.g., a heat carrier fluid. The coolant cooler allows to keep the use of coolant low by cooling it down. Further, the recuperated heat may be used as process heat, e.g., to dry or preheat the raw clay and/or to (pre-) heat a steam boiler and/or to drive an ORC-process (Organic Rankine Cycle process). In short, the cooler can contribute to reduce CO.sub.2 emissions as well as to reduce operating costs. Generalizing the coolant cooler may include a cold heat carrier fluid inlet and a hot heat carrier fluid outlet. The hot heat carrier fluid outlet of the coolant cooler may preferably be connected to a heat inlet of a heat sink by a hot heat carrier fluid conduit. Similarly, the cold heat carrier fluid inlet of the coolant cooler may preferably be connected to a cold heat carrier fluid outlet of the heat sink via a cold heat carrier fluid conduit. Examples of these heat sinks include, but are not limited to, raw meal dryers, boilers (e.g., steam boilers, ORC-boilers), district heating apparatus, combustion air preheaters, oxygen preheaters, chemical reactors, etc.

[0024] The coolant cooler may be omitted, as well: For example, the coolant may be air or another oxygen comprising fluid. The oxygen comprising fluid may be heated up while passing the coolant channel and may subsequently be used as oxygen source for the calcined clay kiln or any other furnace or kiln to thereby increase efficiency of the corresponding plant. Only to avoid ambiguities, the calcined clay cooler may as well include a coolant cooler and at the same time an oxygen comprising fluid may be used as coolant. The oxygen comprising fluid may, but does not need to, be fed to a furnace or another reactor as oxygen source and the heat being transported by the oxygen comprising fluid can hence be used as process heat. In this example, at least one hot coolant line connects at least one coolant outlet of at least one coolant channel with a reactant inlet of a reactor configured for reacting at least a portion of the coolant. A vivid example for the reactor may be a furnace, e.g., the clay furnace. In another example, a portion of the heated coolant may be cooled down in a coolant cooler and another portion may be provided to the furnace or another kind of reactor. In yet another example, at least a portion of the heated coolant may be cooled down in a coolant cooler and at least a portion of the cooled coolant may be provided from the coolant cooler's coolant outlet to the reactant inlet of the reactor (e.g., of the furnace). All these examples can be combined.

[0025] In a preferred example, one of the two connections may be provided by the coolant line. For example, the cold coolant outlet may be connected with the at least one channel inlet via the coolant line. Alternatively, the hot coolant inlet may be connected with the at least one channel outlet via the coolant line.

[0026] If the calcined clay conduit includes agitation means and/or conveying means extending from the inner side of the conduit wall into the conduit volume, cooling can be enhanced by mixing the calcined clay. Further, conveying of the calcined clay can be improved. For example, protrusions extending from the inner surface of the conduit wall into the conduit volume may agitate the calcined clay while the conduit rotates. By inclining a surface of the protrusion relative to the longitudinal direction of the conduit conveying may be controlled. For example, inclining the surface of the protrusion in the conveying direction may increase the flow of the calcined clay towards the clay outlet at a given rotational speed and horizontal orientation of the conduit, while inclining the surface of the protrusion against the direct conveying direction may reduce the average speed of the calcined clay towards the clay outlet. Both alternatives may be used to adapt the time the calcined clay may be cooled in the cooler and hence the temperature it has at the clay outlet.

[0027] The calcined clay cooler may further include one (or more) support wheel(s) with at least one through hole. An example for a support wheel is a support disc, wherein the disc, different from its mathematical definition, shall imply a circular (preferably flat) structure with a finite thickness. The disc may have through holes or be reduced to spokes or the like.

[0028] The at least one channel wall and the at least one conduit wall may extend through the through-hole in the support wheel(s). At least the at least one channel wall may be radially supported by the support wheel. In a preferred example the calcined clay cooler has a number of support wheels. These may preferably be spaced along the rotational axis. Each of the support wheels may contribute to maintain the conduit wall(s) and/or the channel wall(s) and/or the coolant line(s) in a predefined azimuthal position and in a predefined radial position relative to each other. In a simple example, the support wheel may be a disc with a number of through holes, wherein at least some of the through holes have rims that may support a conduit wall and/or a channel wall and/or the coolant line or any other part that extends along the rotational axis and shall rotate together with the clay conduit(s) around the rotational axis.

[0029] In case the calcined clay cooler includes two or more support wheels a support wall may be attached to the at least two support wheels to enclose at least a portion of the rotating part of the calcined clay cooler. The support wall provides for additional stability, protects the coolant channel(s) and the clay conduit(s) from damage and enhances safety of workers present in the vicinity of the calcined clay cooler. Further, heat losses are further reduced.

[0030] The term connected implies a functional connection. For example, if an inlet of a first duct may be connected to an outlet of a channel, the connection provides for a fluid communication. Each connection mentioned herein can be either a direct or an indirect connection. This means that “connected” can be replaced herein by “directly and/or indirectly connected”. Two pieces are in contact if they touch each other. For example, two plain bearing surfaces being in contact to each other allow for a movement (e.g. a translation or a rotation) of the corresponding parts. The term attachment indicates a mechanical fixation. Two pieces are attached to each other if one is affixed to the other.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] In the following, the invention will be described by way of example, without limitation of the general inventive concept, on examples of embodiment and with reference to the drawings.

[0032] FIG. **1** shows a sketch of a calcined clay manufacturing line with a calcined clay cooler.

[0033] FIG. **2** shows a side view of the rotating portion of the calcined clay cooler.

[0034] FIG. **3** shows a partially sectioned view of the rotating portion of the calcined clay cooler.

[0035] FIG. **4** shows a partially mounted rotating portion of the calcined clay cooler.

[0036] FIG. **5** shows a cross sectional view of the cooler along section plane C-C in FIG. **2**.

[0037] FIG. **6** shows a cross sectional view of the cooler along section plane B-B in FIG. **2**.

[0038] FIG. **7** shows a detail of a calcined clay conduit in a sectional view.

[0039] FIG. **8** shows another detail of a calcined clay conduit in a sectional view.

[0040] Generally, the drawings are not to scale. Like elements and components are referred to by like labels and numerals. For the simplicity of illustrations, not all elements and components depicted and labeled in one drawing are necessarily labels in another drawing even if these elements and components appear in such other drawing.

[0041] While various modifications and alternative forms, of implementation of the idea of the invention are within the scope of the invention, specific embodiments thereof are shown by way of example in the drawings and are described below in detail. It should be understood, however, that the drawings and related detailed description are not intended to limit the implementation of the idea of the invention to the particular form disclosed in this application, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

DETAILED DESCRIPTION

[0042] FIG. **1** shows a calcined clay manufacturing line. Clay may be provided to a clay preprocessing station **10** e.g., by trucks or any other conveying means. In the preprocessing station **10** the clay maybe ground, blended and/or dried. From the clay preprocessing station **10** the preprocessed clay may be provided via an optional preheater **20** to a heating device **30** (e.g., to a rotary kiln **30**) configured for heating the clay up to a predefined temperature. The kiln outlet **39** (more generally the heating device's outlet) may be connected to the upper end of a connection chute **105**. Via the connection chute **105**, calcined clay may be provided to an inlet section **110** of a calcined clay cooler **100**. As can be seen, the optional inlet section **110** may include an inlet duct with a duct inlet. Further, the inlet section may include a number of duct outlets. The number of duct outlets corresponds to (and, in a specific non-limiting case, equals) the number of clay conduits. Generally, the duct outlets are connected to the clay inlets of the clay conduits, thereby enabling the distribution, by the inlet section, of calcined clay from duct inlet via the duct outlets to the clay inlets. In a specific case, each of the duct outlets is connected to the clay inlet of the corresponding clay conduit. In the depicted example, the inlet section may be cylindric (at least essentially), although other shapes are possible as well. However, the depicted cylindric shape provides for an improved homogenization of the calcined clay being subsequently charged to clay conduits **160**. This homogenization results in improved heat recuperation and reduced power requirement for rotating the cooler. In addition, conditioning, e.g., with a water and/or steam lance in the inlet section is more efficient, as the dwell time is maximized.

[0043] In the calcined clay cooler **100**, the calcined clay may be cooled down while in turn a coolant may be heated up. An example coolant may be water, but other coolants may be used as well. The heated coolant may be provided to a coolant cooler to cool the coolant down again and to thereby recuperate the heat previously withdrawn from the calcined clay. For example, this heat may be used for example in the preprocessing station **10**, e.g., for drying the raw clay and/or in the preheater **20** and/or a power plant. As can be seen in FIG. **1**, the calcined clay cooler **100** may have a static portion **101** rotatably supporting a rotating portion **120**.

[0044] As can be seen in FIGS. **2** and **3**, the rotating portion **120** may have support walls **125**. On the support wall **125** may be at least one support ring **126** with a rotationally invariant peripheral surface. These support rings **126** may as well be referred to as tire rings **126** or simply as tire **126**. The tires **126** are configured for rotatably supporting the rotating portion **120**, e.g., on rollers of the static portion **101**. The rotational axis is indicated by reference numeral **102**. Inside the support wall **125** are preferably a number of support wheels **140** being spaced along the longitudinal axis **102** (cf. FIG. **4**). The number **126** may be only one (1) but may as well be bigger, depending on the size of the calcined clay cooler. The support wheels **140** may have through holes being delimited by rims, i.e., by a surface delimiting the corresponding through hole. These rims may support a number of n clay conduits **160** with a conduit wall **162** (shown are n=4 clay conduits **160**, but other numbers are possible as well). Each conduit wall **162** encloses a separate conduit volume **168** (see FIGS. **6** and **7**). As apparent from FIGS. **3** to **7**, the conduits **160** are arranged symmetrically with respect to the rotational axis **102**. In the example, are four conduits **160** (n=4) are shown, but other numbers are possible as well. The centers of the conduits **160** are spaced by $2\pi/4=\pi/2=90°$, and hence a rotation of the conduits by $\pi/2$ maps the centers of a conduit on the center of the neighbored conduit. In the example, the conduits **160** have at least essentially the same shape and hence each rotation by $\pi/2$ maps the contours of the conduit walls **162** onto each other. At this point it is noted that the number n of conduits **160** is only an example and may be altered as set out above. In this case, the azimuthal offset between the centers of the conduits may preferably be adapted accordingly. Again, it is noted, that in practice deviations from a perfect discrete rotationally symmetric azimuthal spacing of the conduits **160** may have to be accepted as indicated above. In the depicted example, the centers of the clay conduits **160** have the same radial distance to the rotational axis, but again, it is noted that this is only a preferred example. As apparent from FIGS. **1** to **3**, the inlet section **110** and the clay conduits **160** are preferably attached to each other, at least indirectly, for example via at least one of the support shell **125** and a support wheel **140**. The attachment of the inlet section **110** to the clay conduits **160** ensuring that the inlet section rotates in phase with the clay conduits **160**.

[0045] In the present example, each clay conduit **160** has a coolant channel **180** with a channel wall **182** surrounding the conduit wall **162** (see detail in FIGS. **6** to **8**). The space between the peripheral surface **164** of the conduit wall and the inner surface **186** of the channel wall **182** may be a coolant channel volume **188** (see FIG. **8**) through which the coolant may flow from a channel inlet to a channel outlet. Protrusions **170** may extend inwardly from the inner surface **164** of the conduit wall **162**. In operation, the optional protrusions **170** improve the heat transfer from the hot calcined clay to the conduit wall **162** and at the same time agitate the calcined clay. In FIG. **3** only a single optional protrusion **170** per clay conduit is shown to declutter the figure. Any other number is possible as well.

[0046] As can be seen in FIGS. **1** to **7**, a coolant line **190** may extend along the rotational axis **102** and may be connected for example by connection pipes **195** to coolant inlets of the coolant channels **180** (see FIG. **6**). From there, the coolant may flow parallel to the effective conveying direction of the calcined clay to the downstream end **109** (see FIGS. **2** and **3**) of the calcined clay cooler **100**. From the downstream end **109** it may flow to a coolant cooler and after being cooled down, back into the coolant line **190**.

[0047] FIG. **8** shows a detail of a clay conduit **160** with a conduit wall **162** being surrounded by a channel wall **182**. Attached to an inner surface **186** of the channel wall **182** may be a first leg **210** of a profile **200**. The optional profile **200** may include said first leg **210** and a second leg **220** with a middle leg **230** in between of the first leg **210** and the second leg **220**. A plain bearing surface **228** of the second leg **220** may contact the peripheral surface **164** of the conduit wall **162** and hence support the conduit wall **162**. The profile **200** may thus support the conduit wall radially, while providing a linear plain bearing enabling the conduit wall **162** and channel wall **182** to expand differently in the axial direction. In addition, the profile **200** by its elasticity may as well compensate for different radial expansions or contractions of the conduit wall **162** and channel wall **182**, which may occur when ramping a calcined clay plant up or shutting the calcined clay plant down, respectively. The peripheral surface **164** of the conduit wall **162** may have at least one block attached to it that limit an axial movement of the plain bearing surface **228** e.g., in the azimuthal direction and/or in the axial direction. In another example, the first leg **210** may be attached to the peripheral surface **164** of the conduit wall **162** and the second leg's plain bearing surface may anti-radially support the channel wall **182**. As shown in FIG. **8**, the longitudinal extension of the first leg **210** and/or the of the second leg **220** may be essentially parallel to the azimuthal direction (with respect to the conduit axis). Alternatively, at least one of the first leg **210** and the second leg **220** may have a longitudinal direction being at least essentially parallel (including antiparallel) to the longitudinal axis and/or the rotational axis.

[0048] It will be appreciated to those skilled in the art having the benefit of this disclosure that this invention is believed to provide a calcined clay cooler. Further modifications and alternative embodiments of various aspects of the invention will be apparent to those skilled in the art in view of this description. Accordingly, this description is to be construed as illustrative only and is provided for the purpose of teaching those skilled in the art the general manner of carrying out the invention. It is to be understood that the forms of the invention shown and described herein are to be taken as the presently preferred embodiments. Elements and materials may be substituted for those illustrated and described herein, parts and processes may be reversed, and certain features of the invention may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of this description of the invention. Changes may be made in the elements described herein without departing from the spirit and scope of the invention as described in the following claims.

LIST OF REFERENCE NUMERALS

[0049] **1** calcined clay plant [0050] **10** preprocessing station (optional) [0051] **20** preheater (optional) [0052] **30** calcined clay heater, e.g. rotary kiln (optional) [0053] **39** calcined clay heater outlet, e.g. rotary kiln outlet (optional) [0054] **100** calcined clay cooler [0055] **101** static portion (optional) [0056] **102** rotational axis [0057] **105** connection chute (optional) [0058] **108** upstream end of calcined clay cooler [0059] **109** downstream end of calcined clay cooler [0060] **110** inlet section (optional) [0061] **120** rotatably supported portion/rotating portion [0062] **125** support wall/support shell (optional) [0063] **126** support rings/tire rings/tires [0064] **140** support wheel (optional) [0065] **160** clay conduit [0066] **161** clay inlet of clay conduit [0067] **162** conduit wall [0068] **164** peripheral surface conduit wall [0069] **166** inner surface of conduit wall [0070] **168** conduit volume [0071] **169** clay outlet of clay conduit [0072] **170** protrusions/agitation means (optional) [0073] **180** coolant channel (optional) [0074] **182** channel wall (optional) [0075] **186** inner surface of channel wall [0076] **188** coolant channel volume [0077] **190** coolant line (optional) [0078] **195** connection pipe (optional) [0079] **200** support profile (optional) [0080] **210** first leg (optional) [0081] **220** second leg (optional) [0082] **228** plain bearing surface (optional) [0083] **230** middle leg (optional)

Claims

1.-**15**. (canceled)
16. A calcined clay cooler comprising an axis of rotation and at least one clay conduit, wherein: each of the at least one clay conduit has a clay conduit wall, each of present clay conduit walls encloses a corresponding clay conduit volume, each of present clay conduit volumes has a clay inlet and a clay outlet, wherein the clay inlet is located upstream from the clay outlet, each of the at least one clay conduit is rotatably supported to rotate relative to the axis of rotation, each of the present clay conduit walls comprises at least one coolant channel with a coolant channel wall enclosing a coolant channel volume and defining a coolant channel inlet and a coolant channel outlet, the calcined clay cooler further comprises an inlet section, wherein the inlet section comprises an inlet duct with a duct inlet and multiple duct outlets, wherein a number of duct outlets corresponds to a number of clay conduits and the duct outlets are connected to the clay inlets of the clay conduits, the calcined clay cooler further comprises a connection chute with a connection chute end, wherein the connection chute end extends into the inlet duct thereby defining a ring gap between the connection chute and the inlet duct, and the ring gap is sealed by: a gasket, and/or at least one fluid line has at least one nozzle opening that faces toward the ring gap and/or that ends in the ring gap and/or in an enclosure of the ring gap.
17. The calcined clay cooler of claim 16, wherein the at least one clay conduit comprises at least two clay conduits arranged symmetrically with respect to the axis of rotation.
18. The calcined clay cooler of claim 16, wherein the coolant channel volume is limited by a corresponding clay conduit wall of the present clay conduit walls and by a corresponding coolant channel wall, wherein the corresponding coolant channel wall encloses at least a portion of the corresponding clay conduit wall.
19. The calcined clay cooler of claim 18, wherein the coolant channel wall supports the conduit wall radially, anti-radially, and floatingly in the axial direction.
20. The calcined clay cooler of claim 16, further comprising at least one support profile with a first leg, a middle leg, and a second leg, wherein: the middle leg is in between the first leg and the second leg and defines an offset between the first leg and the second leg, the first leg is attached to one of a peripheral surface of the conduit wall and an inner surface of the coolant channel wall, the second leg has a plain bearing surface contacting the other of the peripheral surface of the conduit wall and the inner surface of the coolant channel wall, and the offset is measured radially with respect to a longitudinal axis of a conduit axis defined by the clay conduit wall.
21. The calcined clay cooler of claim 20, wherein the first leg of the at least one support profile is attached by welding, brazing, riveting, or gluing, or is unitary with a wall to which it is attached.
22. The calcined clay cooler of claim 16, further comprising a coolant line that extends along the axis of rotation and/or that has a connection port in a downstream half of the calcined clay cooler and/or that is fluidly connected with an upstream end of at least one coolant channel.
23. The calcined clay cooler of claim 16, further comprising a coolant cooler with a hot coolant inlet and a cold coolant outlet, wherein a coolant channel inlet of a given coolant channel wall is connected with the cold coolant outlet and a coolant channel outlet of said given channel wall is connected with the hot coolant inlet.
24. The calcined clay cooler of claim 23, further comprising a cold heat carrier fluid inlet and a hot heat carrier fluid outlet, wherein the hot heat carrier fluid outlet of the coolant cooler is connected to a heat inlet of a heat sink by a hot heat carrier fluid conduit, and wherein the cold heat carrier fluid inlet of the coolant cooler is connected to a cold heat carrier fluid outlet of the heat sink via a cold heat carrier fluid conduit.
25. The calcined clay cooler of claim 23 wherein the cold coolant outlet is connected with at least one coolant channel inlet via the coolant line and/or wherein the hot coolant inlet is connected with at least one coolant channel outlet via the coolant line.
26. The calcined clay cooler of claim 16, wherein: the clay conduit comprises (i) agitation means

and/or (ii) conveying means extending from the inner surface of the conduit wall into the conduit volume, and/or the calcined clay cooler further comprises a support wheel, wherein at least one coolant channel wall and at least one clay conduit wall extend through through-holes in the support wheel, wherein at least one of (i) the at least one of the coolant channel wall and (ii) the at least one clay conduit wall is radially supported by the support wheel.
27. The calcined clay cooler of claim 16, further comprising at least two support wheels, wherein at least one coolant channel wall and at least one clay conduit wall extend through through-holes in a support wheel of the at least two support wheels, and wherein a support wall is attached to the at least two support wheels.
28. A calcined clay cooler comprising an axis of rotation and at least one clay conduit, wherein: each of the at least one clay conduit has a clay conduit wall, each of present clay conduit walls encloses a corresponding clay conduit volume, each of present clay conduit volumes has a clay inlet and a clay outlet located downstream from the clay inlet, each of the at least one clay conduit is rotatably supported to rotate relative to the axis of rotation, each of clay conduit walls of the present clay conduit walls comprises at least one coolant channel with a coolant channel wall enclosing a coolant channel volume and defining a coolant channel inlet and a coolant channel outlet, the coolant channel volume is limited by the clay conduit wall and by the coolant channel wall, wherein the coolant channel wall encloses at least a portion of the clay conduit wall, and the coolant channel wall supports the clay conduit wall radially, anti-radially, and floatingly in the axial direction.
29. The calcined clay cooler of claim 28, wherein: the at least one clay conduit comprises at least two clay conduits arranged symmetrically with respect to the axis of rotation, and/or the calcined clay cooler further comprises an inlet section, wherein the inlet section comprises an inlet duct with a duct inlet and a number of duct outlets, wherein the number of duct outlets corresponds to a number of clay conduits and wherein each of duct outlets is connected to the clay inlet of the corresponding clay conduit.
30. The calcined clay cooler of claim 28, wherein: the calcined clay cooler further comprises an inlet section, wherein the inlet section comprises an inlet duct with a duct inlet and a number of duct outlets, wherein the number of duct outlets corresponds to a number of clay conduits and wherein a number of duct outlets corresponds to a number of clay conduits and the duct outlets are connected to the clay inlets of the clay conduits, the calcined clay cooler further comprises a connection chute with a connection chute end that extends into the inlet duct thereby defining a ring gap between the connection chute and the inlet duct, and the ring gap is sealed by (i) a gasket and/or (ii) at least one fluid line having at least one nozzle opening that faces toward the ring gap and/or that ends in the ring gap and/or in an enclosure of the ring gap.
31. The calcined clay cooler of claim 28, further comprising at least one support profile with a first leg, a middle leg, and a second leg, wherein: the middle leg is in between the first leg and the second leg and defines an offset between the first leg and the second leg, the first leg is attached to one of a peripheral surface of the conduit wall and an inner surface of the channel wall, the second leg has a plain bearing surface contacting the other of the peripheral surface of the conduit wall and the inner surface of the channel wall, and the offset is measured radially with respect to a longitudinal axis of a conduit axis defined by the clay conduit wall.
32. The calcined clay cooler of claim 31, wherein the first leg of the at least one support profile is attached by welding, brazing, riveting, or gluing or is unitary with a wall to which it is attached.
33. The calcined clay cooler of claim 28, wherein the calcined clay cooler further comprises: (**33**A) a coolant line that extends along the axis of rotation and/or that has a connection port in a downstream half of the calcined clay cooler and/or that is fluidly connected with an upstream end of at least one coolant channel, and/or a coolant cooler with a hot coolant inlet and a cold coolant outlet, wherein a coolant channel inlet is connected with the cold coolant outlet and a coolant channel outlet is connected with the hot coolant inlet, or (**33**B) a coolant cooler with a hot coolant

inlet and a cold coolant outlet, wherein a coolant channel inlet of a given coolant channel wall is connected with the cold coolant outlet and a coolant channel outlet of the given channel wall is connected with the hot coolant inlet, and wherein the coolant cooler comprises a cold heat carrier fluid inlet and a hot heat carrier fluid outlet, wherein the hot heat carrier fluid outlet of the coolant cooler is connected to a heat inlet of a heat sink by a hot heat carrier fluid conduit, wherein the cold heat carrier fluid inlet of the coolant cooler is connected to a cold heat carrier fluid outlet of the heat sink via a cold heat carrier fluid conduit, or (**33**C) a coolant cooler with a hot coolant inlet and a cold coolant outlet, wherein a given coolant channel inlet is connected with the cold coolant outlet via a coolant line and a given channel outlet is connected with the hot coolant inlet via the coolant line, wherein the coolant cooler comprises a cold heat carrier fluid inlet and a hot heat carrier fluid outlet, wherein the hot heat carrier fluid outlet of the coolant cooler is connected to a heat inlet of a heat sink by a hot heat carrier fluid conduit, wherein the cold heat carrier fluid inlet of the coolant cooler is connected to a cold heat carrier fluid outlet of the heat sink via a cold heat carrier fluid conduit.

34. The calcined clay cooler of claim 28, wherein: the clay conduit comprises (i) agitation means and/or (ii) conveying means that extends from the inner surface of the conduit wall into the conduit volume, and/or the calcined clay cooler further comprises a support wheel, wherein at least one coolant channel wall and at least one clay conduit wall extend through through-holes in the support wheel, wherein at least one of (i) the at least one of the coolant channel wall and (ii) the at least one clay conduit wall is is radially supported by the support wheel.

35. The calcined clay cooler of claim 28, further comprising at least two support wheels, wherein at least one coolant channel wall and at least one clay conduit wall extend through through-holes in a support wheel of the at least two support wheels, and wherein a support wall is attached to the at least two support wheels.